Assignment 2: TCP Connection Performance Analysis using Mininet and Wireshark

Foundation assignment for CampusChat, NetAttend, and MiniNetTransfer

Objective

Build a TCP client-server application in Mininet and compare two TCP communication designs: persistent TCP connection and new TCP connection per message. Students must measure response time, throughput, generate graphs, and verify TCP behavior using Wireshark.

Topology and Setup

Use Mininet default topology:

```
h1 ---- s1 ---- h2

h1 = Server
h2 = Client
Server IP = 10.0.0.1
```

Start and verify the topology:

```
sudo mn

nodes
net
pingall
```

Main Task

Write one TCP server and one TCP client. The client must support two modes:

- persistent: connect once, send all messages, then close the connection.
- new_connection: open a new TCP connection for every message, receive reply, then close.

Message Format

Every client message must follow this format:

```
MSG_ID|MESSAGE_SIZE|MESSAGE_DATA
Example: 3|512|AAAAAAAAAAAA...
```

Every server reply must follow this format:

```
ACK|MSG_ID|RECEIVED_SIZE
Example: ACK|3|512
```

Experiment Setup

Run Mininet using the following setting:

```
sudo mn --link tc,bw=5,delay=50ms
```

Test these message sizes:

```
128 bytes
512 bytes
1024 bytes
```

For each message size, send 10 messages in both modes. Total messages:

```
2 modes × 3 message sizes × 10 messages = 60 messages
```

Client Requirements

1. Connect to server IP 10.0.0.1 on port 5000.
2. Support both persistent and new_connection modes.
3. Send 10 messages for each message size.
4. Measure response time for each message.
5. Calculate average response time for each mode and message size.
6. Calculate throughput for each mode and message size.

```
Throughput = Total bytes sent / Total time taken
```

Server Requirements

1. Listen on port 5000.
2. Accept TCP connections.
3. Receive messages and extract message ID and message size.
4. Send ACK for each message.
5. Save logs in server_log.txt.

Each server log entry should contain:

```
timestamp, client_ip, mode, msg_id, received_size, ack_sent
```

Wireshark Requirement

Capture TCP packets while running both modes. Use the Wireshark display filter:

```
tcp.port == 5000
```

Required Wireshark screenshots:

```
persistent_handshake.png
persistent_data_packets.png
persistent_connection_close.png
new_connection_multiple_handshakes.png
```

The screenshots must show:

- Persistent mode: one TCP handshake, many data exchanges, and one connection close.
- New-connection mode: many TCP handshakes and many connection closes.

Required CSV Files

1. result_table.csv

Columns must be exactly:

```
roll_no,name,mode,bandwidth_mbps,delay_ms,message_size_bytes,total_messages,average_response_time_seconds,throughput_bytes_per_second,status
```

There must be exactly 6 data rows:

```
2 modes × 3 message sizes = 6 rows
```

2. message_response_log.csv

Columns must be exactly:

```
roll_no,name,mode,message_size_bytes,message_number,response_time_seconds
```

There must be exactly 60 data rows:

```
2 modes × 3 message sizes × 10 messages = 60 rows
```

Required Graphs

Create a folder named graphs/ and save the following graphs:

Graph file	X-axis	Y-axis / Purpose
mode_vs_response_time.png	Mode	Average response time; compare persistent vs new_connection
message_size_vs_throughput.png	Message size	Throughput; compare 128, 512, and 1024 bytes
message_response_time.png	Message number	Response time for 512-byte messages in both modes

Submission Folder

Submit one zipped folder named:

```
ROLLNO_NAME_TCP_ASSIGNMENT.zip
```

Folder structure:

```
ROLLNO_NAME_TCP_ASSIGNMENT/
├── server.py
├── client.py
├── server_log.txt
├── result_table.csv
├── message_response_log.csv
├── graphs/
│   ├── mode_vs_response_time.png
│   ├── message_size_vs_throughput.png
│   └── message_response_time.png
├── screenshots/
│   ├── nodes.png
│   ├── net.png
│   ├── pingall.png
│   ├── server_output.png
│   ├── client_output.png
│   ├── persistent_handshake.png
│   ├── persistent_data_packets.png
│   ├── persistent_connection_close.png
│   └── new_connection_multiple_handshakes.png
└── report.pdf
```

Report

Submit a short report of 3-4 pages. It must include:

1. Name and roll number.
2. Mininet topology.

3. Explanation of TCP client-server communication.
4. Difference between persistent and new-connection modes.
5. Result table.
6. All three graphs.
7. Wireshark screenshots.
8. Answers to the questions below.

Questions:

- Why does TCP use a three-way handshake?
- Which mode had lower response time?
- Why does new-connection mode have more overhead?
- Which mode is better for CampusChat and why?
- Which mode is suitable for NetAttend and why?
- How does message size affect throughput?
- How did Wireshark help verify TCP behavior?

Marking Scheme

Component	Marks
Correct Mininet setup	10
TCP server and client	15
Persistent connection mode	10
New-connection mode	10
Message-size handling	10
Response time and throughput measurement	10
CSV files	10
Graphs	10
Wireshark analysis	15
Report	10
Total	**100**

Important Notes

- The program must run inside Mininet.
- Do not change CSV column names.
- Graphs must be generated from your own result values.
- Wireshark screenshots are compulsory.
- Students may use resources, but they must understand and explain their code.
- During evaluation, students may be asked to run the program and explain TCP handshake, persistent connection, response time, throughput, and Wireshark captures.